

Customer Segmentation & Prediction

Business Recommendations

Prepared for: Marketing & Retention Strategy Team
Project: Week 11 - Advanced Machine Learning Models
Dataset: customer_churn.csv (500 customers, 9 features)
Methodology: K-Means clustering + Random Forest prediction

Executive Summary

We segmented 500 customers into four behavioural groups using K-Means clustering on tenure, monthly charges, total spend, contract type, payment method, paperless billing, and senior-citizen status. A separate Random Forest churn-prediction model was tuned per segment with GridSearchCV. Findings are translated into retention and growth actions for each segment.

Segment-by-Segment Recommendations

Senior Legacy Non-Paperless

Share of base: 26.0% (130 customers) | Avg tenure: 36 months | Avg monthly: \$108 | Churn rate: 14.0% (base rate 10.6%)

Who: Senior customers still receiving paper bills, moderate monthly spend.

Why they matter: Lowest digital engagement - the highest touch cost to serve.

Recommended actions:

1. Assisted transition to paperless (small credit incentive, walk-through help).
2. Phone-first retention campaigns.
3. Simplified plan structures (reduce decision fatigue).
4. Mailers with large-print value comparison to anchor on.

Young Digital Customers

Share of base: 25.2% (126 customers) | Avg tenure: 38 months | Avg monthly: \$121 | Churn rate: 11.0% (base rate 10.6%)

Who: Non-senior customers using paperless billing - the youngest, most digitally-native group.

Why they matter: Largest growth pool - receptive to cross-sell and tenure-building offers.

Recommended actions:

1. Mobile-first push notifications and gamified milestones.
2. Premium trial offers and freemium-to-paid paths.
3. Co-marketing with brands they already trust.
4. Annual-plan nudges at tenure milestones (6/12/24 months).

Mid-Spend Traditional Users

Share of base: 25.0% (125 customers) | Avg tenure: 37 months | Avg monthly: \$117 | Churn rate: 11.0% (base rate 10.6%)

Who: Non-senior customers still on paper bills with above-average monthly spend.

Why they matter: Higher revenue but disengaged - migrate to digital to cut service cost.

Recommended actions:

1. Paperless migration with a clear environmental + cost-saving pitch.
2. Bundle offers that increase stickiness.
3. Targeted loyalty rewards after the first digital action.
4. Annual plans to lock in spend.

Senior Digital Switchers

Share of base: 23.8% (119 customers) | Avg tenure: 35 months | Avg monthly: \$109 | Churn rate: 6.0% (base rate 10.6%)

Who: Senior customers who have adopted paperless but spend more moderately.

Why they matter: Already digital - cheapest channel to engage - and likely to upgrade spend.

Recommended actions:

1. In-app usage tips to unlock more value (drives upgrades).
2. Trial upgrades to premium tiers.
3. Senior-friendly UX improvements (larger fonts, voice support).
4. Health-and-safety partnerships to keep the value prop relevant.

Model Performance

Each segment was modelled independently with a Random Forest tuned by GridSearchCV (5-fold CV, F1-scored). Performance varies because some segments have very few churn events.

Segment	Accuracy	Precision	Recall	F1	ROC-AUC
Senior Legacy Non-Paperles	1.000	1.000	1.000	1.000	1.000
Senior Digital Switchers	0.933	0.000	0.000	0.000	0.866
Mid-Spend Traditional Users	0.844	0.429	0.750	0.545	0.964
Young Digital Customers	1.000	1.000	1.000	1.000	1.000

Estimated Business Impact

With the segment-specific models we can identify approximately **36** of the 53 at-risk customers (Recall=0.69). If retention campaigns successfully convert half of these, that is roughly **18 customers retained** and an estimated **\$24,545/year** in preserved monthly revenue (18 x \$114 average monthly x 12).

Caveats: predictions are limited by the small churn base (53/500); recall and F1 on minority-class segments should be read alongside class imbalance, not as headline numbers.

Methodology Limitations

1. Only 500 customers with 4 numerical features limits statistical power per segment.
2. K-Means assumes spherical clusters - Ward-linkage hierarchical results may better capture non-spherical shapes.
3. Models are not time-aware; tenure represents snapshot only.
4. Class imbalance in churn (10.6%) inflates accuracy - rely on F1/ROC-AUC for comparison.